EZ-ZOUHRY Mohamed Reda

Final-Year Embedded Systems Engineering Student
Real-Time Systems · PCB Design · FPGA · PFE 2027

Marrakech, Morocco | m.reda.ezzouhry@gmail.com | +212 642 715574
linkedin.com/in/mohamed-reda-ez-zouhry | github.com/bk201-dev

PROFILE

- Final-year Embedded Systems & Control Engineering student with hands-on experience in multilayer PCB design, STM32 firmware, real-time systems, FPGA/RTL and industrial communications. Interested in building reliable hardware and software from architecture to validation. Seeking a final-year (PFE) internship starting February 2027.

EDUCATION

Engineering Degree in Embedded Electronic Systems & Control | ENSA Marrakech **2024 – Present**
- Focus on embedded electronics, control systems, robotics, real-time programming and digital hardware design.

Integrated Preparatory Program | ENSA Safi & ENSA Marrakech **2022 – 2024**
- Advanced mathematics, physics, computer science and engineering fundamentals.

EXPERIENCE

Embedded Systems R&D Intern | ELECT-CAD COMPANY, Casablanca **Jul 2026 – Sep 2026**
- Developed the architecture of an STM32F407 monitoring board for transformer cabinets with isolated RS-485/Modbus, LTE connectivity and local data storage.
- Designed a four-layer PCB in Altium Designer, considering electrical isolation, power integrity, RF routing, protection and EMI/EMC.
- Performed electrical and mechanical design reviews, PCB rule checks and verification of isolation constraints.

Industrial Automation & Electrical Engineering Intern | SRM MS, Marrakech **Jul 2025 – Aug 2025**
- Studied PLC-based control, industrial electrical protection and automated systems.
- Analyzed distribution networks, transformer substations and voltage adaptation/protection equipment.

Open-Source Engineering Projects | GitHub **2024 – Present**
- Documented personal electronics, PCB and embedded-system projects through public repositories and technical READMEs.
- Shared hardware architecture, embedded code and simulation results for reproducibility and learning.

PROJECTS

- **EdgeGuard — Industrial Monitoring Hardware (STM32 / Modbus / LTE) —** Developed a connected industrial monitoring architecture for transformer cabinets, with an STM32F407, isolated RS-485 transceiver, cellular modem, EEPROM and dedicated power/protection circuits. Designed a four-layer PCB considering isolation and EMI/EMC.
- **Mixed-Signal STM32 Controller (Ethernet / Motor Control / ADC) —** Designed a four-layer STM32F407 control board integrating an Ethernet PHY, two motor drivers, ADC/DAC and analog signal conditioning. Worked on RMII, USB 2.0, CAN, SPI, I²C and UART interfaces, stack-up planning and routing constraints.
- **32-bit RISC-V Single-Cycle CPU (SystemVerilog) —** Designed an RV32I processor with instruction fetch/decode, register file, ALU, control logic, instruction/data memory and branch execution. Simulated and verified Fibonacci and array-processing test programs using ModelSim and Quartus.
- **Robotic Arm & Digital Twin (ROS / Gazebo / MoveIt) —** Created a URDF-based 3D robot model and simulated actuation, joint control and motion planning in ROS/Gazebo. Worked on ROS topics, services and actions, and Arduino/rosserial integration for the physical arm.
- **ESP32 IoT Monitoring Platform (FreeRTOS / AWS IoT) —** Built an ESP-IDF application with Wi-Fi access point and station modes, MQTT cloud messaging, OTA update, HTTP monitoring and sensor acquisition. Used FreeRTOS tasks, NVS and SNTP to organize concurrent services.
- **Autonomous Sumo Robot (STM32 / Sensors / PCB) —** Designed and routed a custom robot control PCB around an STM32 Blue Pill, dual TB6612FNG motor drivers, distance and line sensors, IR receiver and protected supply. Developed the sensor and actuator interface architecture.
- **Automotive Multi-Node Network (STM32 / FreeRTOS / CAN / UDS) —** Implemented a distributed STM32 system with CAN frame exchange, real-time processing and gateway communication. Added Bluetooth connectivity and diagnostic behavior based on UDS principles.
- **RC Flying Wing — Multidisciplinary Design (MATLAB / RF / PCB) —** Designing an RC flying wing by combining aircraft aerodynamics, MATLAB analysis, electronic architecture and embedded radio control. Covers actuator and ESC selection, power distribution, sensor interfaces and PCB development.

TECHNOLOGIES & TOOLS

- **Programming & RTL:** C/C++, Python, MATLAB, VHDL, Verilog, SystemVerilog, RV32I Assembly.
- **Electronics & Hardware:** STM32, ESP32, four-layer PCB, mixed-signal design, isolation, EMI/EMC, FPGA (Cyclone V).
- **Software & Engineering Tools:** Altium Designer, KiCad, Quartus, ModelSim, CubeIDE, ROS, Gazebo, Simulink, Git.
- **Embedded Systems & Communication:** FreeRTOS, CAN/UDS, Modbus RTU, RS-485, UART, SPI, I²C, Ethernet, MQTT.

LANGUAGES

- Arabic (Native), French (C1), English (C1).